Preparation and Verification of Two-Qubit Bell State |Phi+>

Objective: Prepare maximally entangled state and verify outcomes.

Hypothesis: Applying H on qubit 0 and CX on qubits 0 and 1 creates Bell state.

Methodology: 2 qubits initialized in |00>. Gates: H on qubit 0, CX on qubits 0 1.

Expected Results: Outcomes |00> with prob 0.50 and |11> with prob 0.50.

Assumptions: Noiseless quantum processor with ideal gate fidelity.